

HARRISON PARK

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EDUCATION

University of Toronto

September 2022 - June 2026

Bachelor's, Physics and Statistics Dual Major; Computer Science Minor

GPA: 3.97

- President's Scholar, awarded \$7000 and Dean's List Scholar for 2023, 2024, and 2025
- Awarded the David and Louis Fraser Scholarship, Department Of Physics Undergraduate Scholarship, New College Council In-course Scholarship, The Donald G. Ivey Scholarship In Physics, The Harry Boxen Memorial Scholarships In Physics, and the William Ramsay Scholarship

PROFESSIONAL EXPERIENCE

Vector Institute

May 2026 - Present

Artificial Intelligence Research Intern

Toronto, ON, Canada

- As a guest researcher at the National Artificial Intelligence Institute in Toronto under the Pan-Canadian Artificial Intelligence Strategy, my work focuses on improved inference for scaling laws for Large Language Models
- Investigate better optimizers and loss functions for fitting the Chinchilla scaling model to better scale data from LLM training.

PROJECTS

Improving Scaling Laws for Large Language Models

May 2026 - Present

Toronto, ON, Canada

- Investigate strategies for optimizing loss functions building on the Chinchilla Approach 3 power law loss objective for Large Language Models
- Work with AI-dedicated GPU clusters hosted as part of the Pan-Canadian AI Compute Environment
- My project is self-directed and scoped, with oversight under the supervision of Dr. Chris Maddison

Steam Game Review Sentiment Analysis using ML

- Developed an interactive dashboard to analyze Steam game reviews using DistilBERT for sentiment classification.
- Built visualizations with Seaborn, Altair, and wordcloud; deployed the dashboard using Streamlit Cloud.
- Tech stack: Python, pandas, scikit-learn, NLTK, Hugging Face Transformers, Streamlit.

Cocoa Price Time Series Forecasting

- Developed skills relevant to financial and economic time series forecasting, predicted 4 months future prices with MAPE 7%
- Analyzed cross-correlations with local climates, CPI, and exchange rates; utilized SARIMAX, regression, GARCH, and recurrent neural network models

RELEVANT COURSEWORK / EXPERIENCES

STA414: Statistical Methods for Machine Learning II

- Developed skills in ML workflows and theory, proficient in both regression and classification problems. Models/techniques include Gaussian Processes, Variational Inference, Neural Networks, EM, VAE, HMM, MCMC, Transformer Architectures, etc.

STA457: Time Series Analysis

- Developed proficiency with various time series models, including SARIMAX, GARCH, and ETS, in R.

PHY407: Computational Physics

- Improved skills in data modelling and numerical integration, Fourier analysis, PDE solving, and Monte Carlo techniques, in Python.

PHY408: Time Series Analysis

- Gained further data modelling skills such as spectral analysis, physical modelling, convolutions, filtering, and interpretation.

CSC401: Natural Language Computing

- Developed proficiency with many NLP techniques, including Information Theory, Deep NNs, Machine Translation, Transformers, LLMs, Speech Recognition, IR, RAG, etc.

SKILLS

Skills: Python, Data Analysis, R, Modelling, Tableau, SQL, Git, Java, Time Series, Machine Learning, NLP, AI